Training Purpose and Logistics

Purpose and expectations

- The five-day Data Excellence Training prepares participants to turn business needs into trusted, reusable data products that support business priorities at scale.
- The program is hands-on and intended for entry- and intermediate-level participants. Participants were encouraged to stay curious, ask questions, share feedback, and apply learning to real work.
- The program is evolving; participant feedback is intended to inform improvements.
- Participants may step out briefly for occasional conflicts if this is not disruptive. Organizers should be notified about extended absences.
- Lunch voucher information was provided for the week.

Schedule and course plan

- Course hours: 9 a.m. to 3 p.m., with lunch at noon and morning and afternoon breaks.
- The Friday capstone is expected to finish around 2 p.m. if participants stay focused.

Monday: Data-product design, mindset, soft skills, group discussion, and AI practice.

Tuesday: The paved-path technology stack, including Docker and Git.

Wednesday: DBT for building data products.

Thursday: Airflow for orchestration, scheduling, and SLA management.

Friday: Capstone implementation and certification.

- Databricks, harnesses, deployment practices, and related tools may be discussed alongside the main topics.

Facilitators

- **Michael Burgess** is Head of Technical Consulting at Decoded, an international skills consultancy. His background includes software engineering, defense, work connected to the 2012 Olympics, and education. He works with clients and subject-matter experts on training-program design.

- **Sophie McMurray** has nine years of experience in data-focused roles, a background in mathematics and statistics, and a master's degree in data science. She works primarily on AI engineering programs and supports this training.

- **Hero** introduced Capital Group's data-excellence operating model and the reasons for the training.

Capstone

- The capstone involves creating a data product from provided CSV data concerning funds and related information, deploying it to Caspian Databricks, and demonstrating the implementation.
- The first phase is an observed certification checkpoint.

Data Excellence Operating Model

Why the model is needed

- The goal is self-service access to trusted firm-wide data that people and AI can discover and use quickly and confidently.

- Access must be governed: users must be authorized, aggregation policies followed, and personally identifiable information (PII) protected.
- Technologists may know how to query tables without knowing the business context, authoritative sources, or policies required to decide how data should be linked, shared, or used.
- Leaving business decisions solely to technologists can create inconsistent assumptions and unfair accountability.

Four persona archetypes

- **Consumers**: Anyone who needs data, including business partners, managers, and technologists.

- **Platform providers**: Establish approved standards, tools, policies, catalogs, and metadata for creating and managing data.

- **Business data owners**: Senior business stakeholders accountable for data in their domains and for decisions about its use and sharing.

- **Data product creators**: Build reusable products by bringing together platform standards, owner priorities, and consumer needs.
- Business-owner domains mentioned include client data, results, positions, attribution, trades, and research data. The DXP site has a directory of data owners and stewards.
- Product creators work at the intersection of the other personas: they need to understand consumer needs, owner priorities, and how to use the platform correctly.

Paved path and discoverability

- The platform-provider model is intended to replace uncertainty about which platform or tools to use with a consistent, policy-compliant way to create data.

- Tools mentioned included Laser, Caspian, EMR, Databricks, and Dremio.
- Correct catalogs and useful metadata make products easier for people and AI to find.
- Publishing products with appropriate metadata and access controls lets consumers find data in a catalog rather than relying on an individual to provide it.

Data Products

Definition and perspectives

- A data product can be understood as a dataset accompanied by promises about its use and quality. Common properties include:
  - Trusted
  - Reusable
  - Business-ready
  - Accessible
- Promises can be documented in design materials and contracts, then enforced through automated quality checks and code running on supporting infrastructure.
- From an engineering perspective, a product includes the data, transformations, code, and infrastructure that implement its promises.
- From a consumer or product perspective, the focus is on the usefulness of the data and the needs it serves.
- Merely placing a third-party dataset, such as Morningstar data, in a folder does not by itself make it a Capital Group data product.

Consumer alignment and value

- A data product transforms source-aligned data into consumer-aligned data.
- Source-aligned data reflects the provider’s interests and structure. Consumer-

aligned data is shaped around the needs and objectives of its intended users.
- Alignment is relative to people's goals and needs; it is not a technical property that can be established solely through code or automated tests.
- Much of a data product's value comes from transforming data to meet consumer needs while making it dependable and accessible.
- Engineering supports both realignment through transformations and business rules, and the enforcement of promises such as quality standards and SLAs.

What a data product is not

- It is not merely a source dataset.
- It is not a consumer-aligned dataset left without supporting processes.
- It is not code and infrastructure without meaningful transformation, consumer alignment, and enforced promises.

Early-Stage Data-Product Design

Initial discovery and triage

- Clarify who needs the product, its business purpose, whether it already exists, what people do now (including workarounds), and whether it is still needed.
- Assess whether and how it can be built, including whether the data exists, its sources, required transformations, platform, data grain, sensitivity controls, delivery method, update frequency, adoption, and timing.
- Consider why consumers would use the product instead of existing options.

Before deciding to build, consider viability, value, cost, and appropriate prioritization. The facilitator described this early decision-making as the **triage** stage.

Business case: costs and benefits

- Costs discussed included:

    - Total cost of ownership and maintenance.
    - Development time and team capacity.
    - Opportunity costs: work that cannot be done while teams build or maintain the product.
    - Data licensing and acquisition.
    - Subject-matter expert and data-owner time.
    - Coordination and alignment time.
    - Adoption costs, such as training and communications.
- Benefits discussed included:
    - Less time spent by downstream consumers.
    - Greater consistency and less rework or coordination.
    - New business opportunities and better opportunities for insight when related data is brought together.
    - Greater trust, reduced friction, and more scalable operations.
    - Lower transaction costs from shared standards and tools.
    - Better support for AI workflows and process efficiency.
- Difficult-to-price benefits can be estimated using proxies, such as meetings, handoffs, or alignment work that a product could reduce.
- Benefits closer to revenue-generating work can be easier to quantify. Leadership may need to make informed judgments about benefits that are real but difficult to measure precisely.

Adoption

- Lack of adoption can cause a product to fail even if it has technical merit.
- Barriers discussed included:
    - Difficulty integrating with existing systems.
    - Low trust in data quality or expected benefits.
    - Unclear liability or uncertainty about whether use is authorized.
    - Lack of awareness, training, or documentation.

    - Competing priorities and the effort of learning a new system.
    - Resistance to change or preference for familiar practices.
    - Products that do not support the actual use case.
    - Rigid API or data contracts that cannot accommodate different consumers.
- Leadership can reduce uncertainty, clarify priorities, back experimentation, and communicate responsibility.
- Demonstrations, training, documentation, communications, internal announcements, and recorded walkthroughs can help consumers see value and learn to use a product.
- Deadlines may help resolve competing priorities. Consistent communication and follow-through help maintain trust.

Interviews and Requirements

Ways to understand consumer needs

- Methods discussed included interviewing consumers, observing their current work, reviewing past behavior and documentation, using surveys, and examining relevant code or other work artifacts.
- Interviews support dynamic follow-up and discovery. Surveys may be less effective when questions reflect the creator’s assumptions rather than consumers’ needs.
- Observation can reveal actual workarounds, needs people may not be able to articulate, or differences between what they say and what they do.

Interview prompts

- What decision do you make with the data?
- How do you make that decision today?
- Who uses the result directly?
- How is the source generated?

- What do the relevant terms mean in your work?
- How do you recognize that the result is usable?
- When do you read the result?

Requirements documents

- A **product requirements document (PRD)** describes the product's purpose and requirements.
- A **data contract** gives more specific expectations for data, such as identity, schema, SLAs, data quality, dependencies, glossary, and change management.
- A glossary is useful in a data contract because teams and sources may use the same terms differently; upstream providers may define their own terminology.
- Requirements should reflect consumer and producer input. AI-generated documents can help draft artifacts but do not replace stakeholder understanding.

Group Examples and Challenges

Data products mentioned

- Client data platforms that combine data from different client domains.
- Positions and transactions, including positions sourced from an accounting book of record.
- Website and marketing engagement data.
- Enterprise-search data held on a vendor platform.
- Investment attribution, product reference data, indexes, and portfolio data.
- Client and advisor profiles, including matching, mastering, and deduplication.
- Assets and client sales flows.
- Advisor data distributed across the enterprise through APIs and files.

Common data and coordination issues

- Finding authoritative sources and determining the correct business logic and keys for linking datasets.
- Aligning teams that own different domains and resolving discrepancies between sources.
- Reconciling multiple profiles for the same person or client.
- Integrating data held in silos and packaging it for consumers.
- Assessing the correctness of external data, which may require comparison with other datasets.
- Examples of data-quality concerns included implausible sales figures that should be compared with assets under management, data-entry errors such as an extra zero, conflicting sources, and uncertainty about whether information came from the appropriate sender.
- As organizations grow, the number of teams, owners, systems, and data lineages can increase alignment and transaction costs.

Advisor data and CDM

- Advisor data is used by contact centers, digital websites, CRM, marketing, sales, wholesalers, client services, compliance, and sales finance.
- Sales users use advisor information to build relationships; leaders use dashboards to understand interactions and business development.
- Consumers may need different dimensions: marketing may focus on names and contact details, while other groups may need affiliations, trading identities, or product information.
- Rigid API or data contracts can make it difficult to satisfy different consumer requirements.
- A related CDM discussion focused on maintaining one trusted, current record for every advisor office, including who it is, whether it is active, and how it can be covered.
- Globalization creates challenges when regional operations have different levels

of maturity, data standards, structures, and regulatory requirements. These issues can affect feasibility, design, and adoption.

Marketing engagement and client data

- A group discussed personalized marketing data and measuring engagement across channels such as email and search.
- Relevant data is spread across teams and systems, making integration and packaging into a consumable product difficult.
- The group identified a need for clear measurement and a feasible integration path.
- Client-profile discussions included deduplication and determining whether externally sourced data is correct by correlating it with other sources.

Catalyst and AI-Assisted Work

Catalyst skills

- Catalyst was described as a collection of AI skills for work across the data product lifecycle, from design documentation to technical artifacts such as Python and DBT.
- Skills are reusable instructions, often stored as Markdown prompts, that guide the AI to produce artifacts in consistent structures aligned with company policy and standards.
- A data-contract skill can generate a structured document covering identity, schema, SLAs, and data quality.
- Lifecycle skills can call more specific skills in sequence, including requirements and acceptance criteria, feasibility analysis, consumer use cases, and data modeling.
- Some skills include Python scripts that allow operations such as searching Databricks or running jobs.

- The broader DPLC concierge skill covers the full lifecycle and can take a long time to run; focused skills were recommended when appropriate.
- Catalyst setup and access are covered by onboarding materials. A missing repository link was identified in discussion as an access issue.

Lifecycle phases and build requirements

- The session elicited this lifecycle sequence: prerequisites, triage, prioritization, discovery, design, build, validate, publish, operate, and evolve.
- The speaker also referred to the lifecycle as having nine phases, although the spoken sequence contains more than nine labels.
- Build-phase artifacts mentioned included a PRD, data contract, data model, and transformation specification.
- Catalyst can be queried about lifecycle phases, phase requirements, and prerequisites for using particular skills.

AI-generated documentation

- Catalyst can draft PRDs and data contracts from a product description, including information gathered in discussions or transcripts.
- Output varies with the specificity and framing of the input. Comparing independently created documents for the same product can expose missing requirements, differing schemas, and assumptions introduced by the model.
- A PRD may seem more successful than a data contract because it is broader and less schema-specific.
- Generated fields and attributes should be checked against authoritative sources and business requirements.

AI Concepts and Effective Practice

Definitions and framing

- The instructor defined artificial intelligence as a human project of building systems that perform tasks requiring intelligence in people.
- The instructor offered a biological framing of intelligence as a strategy animals use when no known solution is available. This differs from a computer-science framing focused on whether a problem has been solved.

The instructor described current generative AI as a **proxy model**: it can reproduce some properties associated with human capabilities without necessarily possessing those capabilities itself.

- In the instructor's framing, an LLM operates in a text domain: it receives and generates text. Text describing a future or a plan is not, by itself, evidence that the model has simulated that future.

Discriminative and generative AI

- **Discriminative AI** assigns labels, classifications, or predictions to inputs; it answers questions such as "What kind of thing is this?" or "Yes or no?"
- **Generative AI** produces new material resembling examples it has seen. LLMs generate text resembling patterns in their training data.

LLM training and refinement

- The instructor described LLM training data as including internet text, ebooks, and other digitized text.
- Vendor conversation data may provide additional examples of human interactions, depending on vendor practices.

Instruction fine-tuning uses examples pairing instructions with appropriate responses so the model learns to follow instructions rather than merely continue text.

Reinforcement learning from human feedback (RLHF) uses preference

judgments from human reviewers to indicate which of several outputs is better.

- Some tasks, especially code, can be evaluated partly through automated checks such as running code. The instructor noted that “it runs” is a limited quality signal.
- The instructor characterized LLMs as compressing patterns from training data and warned that training on model-generated output can degrade quality. These were presented as explanatory framing during the session.

Inference, reasoning, tools, and agents

- **Training** creates a model; **inference** is the process of running it.
- The instructor explained that some modern systems generate hidden intermediate tokens, often called reasoning, before producing a visible response. This can increase inference cost.
- Output quality may depend on whether training data contains useful examples of the relevant reasoning or steps. Shared explanatory material, such as code examples or technical discussions, can provide such examples.
- A **tool call** is structured text, often JSON, that a program can interpret as an instruction to perform an action.
- A **harness** interprets model tool calls and performs actions. VS Code and Claude Code were given as examples of interfaces or harnesses; Opus was given as an example of a model.
- An **agent** was described as a harness combined with an LLM: the model produces actions and the harness executes them.
- In this framing, AI skills are prompts that can guide an LLM to generate text or trigger tool calls.

Models, context, and effective use

- Model-selector examples included Opus, Sonnet, and Haiku. The instructor described them as differing in size, compression, capability, and cost.

- Model choice is an empirical question: test a model on the actual task. A more capable model may matter less when the prompt supplies extensive context; a less specific prompt may rely more on model capability.
- The “thinking effort” control was described as changing the amount of hidden intermediate-token generation, not human-like thinking. More is not always better; excessive intermediate text may reduce performance on simple tasks.
- Context contains conversation history, including tool calls and hidden reasoning. Performance may degrade as the conversation grows, and systems may compress or summarize older context.
- The instructor recommended starting a fresh conversation for a new task or summarizing work into a file and carrying that summary into a new conversation, rather than relying on automatic compression.
- Effective AI use requires skepticism when a task is novel or poorly represented in training examples. For unfamiliar tasks, compare options and observe actual performance rather than assuming the model will work reliably on the first attempt.
- Cost figures discussed for frontier models were rough instructor estimates, not course commitments: training was described as costing tens of millions of dollars, while running a frontier service was described as costing hundreds of millions per month. The instructor emphasized inference infrastructure and compute as major cost drivers.

Voice-Recorded Interview Exercise

Recording and transcript workflow

- The facilitator demonstrated Apple Voice Memos on a MacBook: record a conversation, pause and finish the recording, view the transcript, and copy it for use in another tool.
- The built-in transcript was described as free and adequate for the exercise, though not perfect.

- The facilitator noted that participants could use an approved app available on their laptops; details of internal voice tools were not known.
- Participants using Zoom or Teams could use the meeting transcript instead.

Interview structure

- The exercise used three roles: **data producer**, **data consumer**, and **data creator**.
- The producer should know the data source well; the consumer should know what they need; the creator asks questions to determine how to shape the product.
- The interview was intended to establish what should be built after the group had considered whether building the product was worthwhile and feasible.
- Suggested questions included:
  - What decision do you make with the data?
  - How do you make that decision today?
  - Who uses the result directly?
  - How is the source generated?
  - What do these terms mean in your work?
  - How do you recognize the result is usable?
  - When do you read the result?
- The facilitator encouraged participants to add questions relevant to understanding the consumer’s needs.
- Groups were asked to record the discussion and use the transcript with Catalyst to draft a PRD and data contract. Participants were encouraged to stay in role but could help one another if the roleplay stalled.